

# Medicine Management System / Pharmacy Management System

## ER Diagram, Flowchart Prompt, and Algorithm Prompt

### 1. ER Diagram Prompt

Create a complete, clear, and simple ER (Entity Relationship) Diagram for a Medicine Management System / Pharmacy Management System.

The system must include the following features:

- Medicine details
- Customer details
- Quantity
- Sale
- Bill calculation

Identify all entities, attributes, Primary Keys (PK), and Foreign Keys (FK).

Use the following entities:

#### **Medicine:**

- Medicine\_ID (PK)
- Medicine\_Name
- Category
- Price
- Stock\_Quantity
- Expiry\_Date

#### **Customer:**

- Customer\_ID (PK)
- Customer\_Name
- Contact\_Number
- Address

#### **Sale:**

- Sale\_ID (PK)

- Customer\_ID (FK)
- Sale\_Date

**Sale\_Item:**

- Sale\_Item\_ID (PK)
- Sale\_ID (FK)
- Medicine\_ID (FK)
- Quantity
- Price
- Item\_Total

**Bill:**

- Bill\_ID (PK)
- Sale\_ID (FK)
- Total\_Amount
- Bill\_Date

Show the relationships clearly:

- One Customer can have many Sales.
- One Sale belongs to one Customer.
- One Sale can contain many Sale\_Items.
- One Medicine can appear in many Sale\_Items.
- One Sale generates one Bill.
- Show proper cardinality such as 1:N and 1:1.

Use standard ER diagram notation with entities, attributes, relationships, PK and FK clearly marked.

Keep the diagram simple, neat, readable, and suitable for a 1st-year BCA project.

Avoid overlapping relationship lines and arrange the entities properly.

## 2. Flowchart Prompt

Create a complete, simple, and clear flowchart for a Medicine Management System / Pharmacy Management System.

The system should perform the following operations:

1. Start
2. Enter customer details
3. Display available medicine details
4. Select medicine
5. Enter quantity
6. Check medicine stock
7. Calculate item total
8. Add item total to the total bill
9. Ask whether the customer wants to purchase another medicine
10. If Yes, select another medicine and repeat the process.
11. If No, calculate the final bill
12. Display customer details
13. Display selected medicine names, prices, and quantities
14. Display final bill amount
15. Complete the sale
16. End

Use the following formulas:

- Item Total = Medicine Price × Quantity
- Total Bill = Sum of All Item Totals

Include a decision box for checking whether sufficient stock is available. If stock is not available, display "Insufficient Stock" and allow the user to select another medicine.

Use standard flowchart symbols:

- Oval for Start/End
- Parallelogram for Input/Output
- Rectangle for Processing
- Diamond for Decision

Make the flowchart neat, simple, properly connected with arrows, and suitable for a 1st-year BCA project.

Do not make the flowchart unnecessarily complicated.

### 3. Algorithm Prompt

Create a simple and complete algorithm for a Medicine Management System / Pharmacy Management System suitable for a 1st-year BCA student.

The system features are:

- Medicine details
- Customer details
- Quantity
- Sale
- Bill calculation

Write the algorithm in simple numbered steps.

Include the following:

**Title: Algorithm for Medicine Management System**

**Input:**

- Customer ID
- Customer Name
- Contact Number
- Medicine ID
- Medicine Name
- Medicine Price
- Quantity

**Processing:**

17. Accept customer details.
18. Display available medicine details.
19. Select a medicine.
20. Enter the required quantity.
21. Check whether sufficient medicine stock is available.
22. If stock is insufficient, display an appropriate message and select another medicine.
23. Calculate the item total using:  $\text{Item Total} = \text{Medicine Price} \times \text{Quantity}$
24. Add the item total to the total bill:  $\text{Total Bill} = \text{Total Bill} + \text{Item Total}$
25. Ask whether the customer wants to purchase another medicine.
26. If Yes, repeat the medicine selection and quantity process.
27. If No, generate the final bill.
28. Update the medicine stock after the sale.
29. Display customer details, selected medicines, quantities, prices, and final bill amount.

**Output:**

- Customer details
- Medicine details
- Quantity purchased
- Individual item totals
- Final total bill
- Sale completed message